Simulation Hybrid Cutoff Heatmaps Report

2026-06-24

1 Overview

This report shows hybrid AE/ARE cutoff heatmaps for each of the five α values.

For a given pair of thresholds (τ_{AE}, τ_{ARE}), the best hybrid cutoff c^* is the value of c that maximises the success rate under the rule: use AE when $\hat{p} \leq c$, otherwise use ARE.

The simulation uses the following settings:

Table 1: Simulation configuration used in this report.

Parameter	Value
alpha	2, 2.5, 3, 4, 5
K	10
n	10000
B	500
cutoffs	seq(0, 0.5, by = 0.001)
tau_AE values	0.005, 0.01, 0.02, 0.03, 0.04
tau_ARE values	0.025, 0.05, 0.1, 0.2, 0.5
maximize	cell
seed	260926

2 True proportions

The true proportions for each α are shown below. Proportions are derived from the weighted Beta(α, 1) quantile grid.

Table 2: True proportion values by alpha and cell-type index.

alpha	index_1	index_2	index_3	index_4	index_5	index_6	index_7	index_8	index_9	index_10
2.0	0.01000	0.03000	0.05000	0.07000	0.09000	0.11000	0.13000	0.15000	0.17000	0.19000
2.5	0.00280	0.01454	0.03129	0.05184	0.07558	0.10212	0.13120	0.16261	0.19620	0.23182
3.0	0.00075	0.00677	0.01880	0.03684	0.06090	0.09098	0.12707	0.16917	0.21729	0.27143
4.0	0.00005	0.00136	0.00628	0.01724	0.03663	0.06688	0.11040	0.16960	0.24688	0.34467
5.0	0.00000	0.00026	0.00197	0.00757	0.02068	0.04614	0.09000	0.15953	0.26319	0.41067

3 Hybrid cutoff heatmaps

Each heatmap shows, for a grid of (τ_{AE}, τ_{ARE}) threshold pairs, the best hybrid cutoff c^*. Cell labels give the cutoff value; colour intensity indicates magnitude.

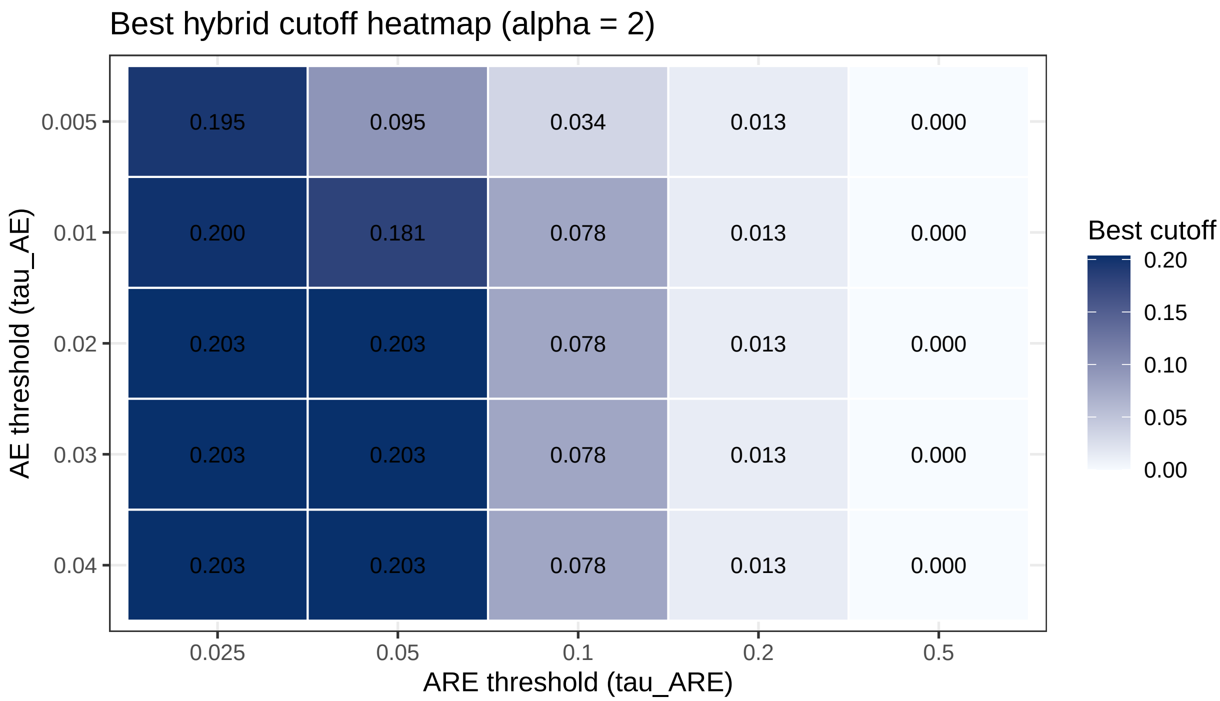

Figure 1: Best hybrid cutoff heatmap for alpha = 2.

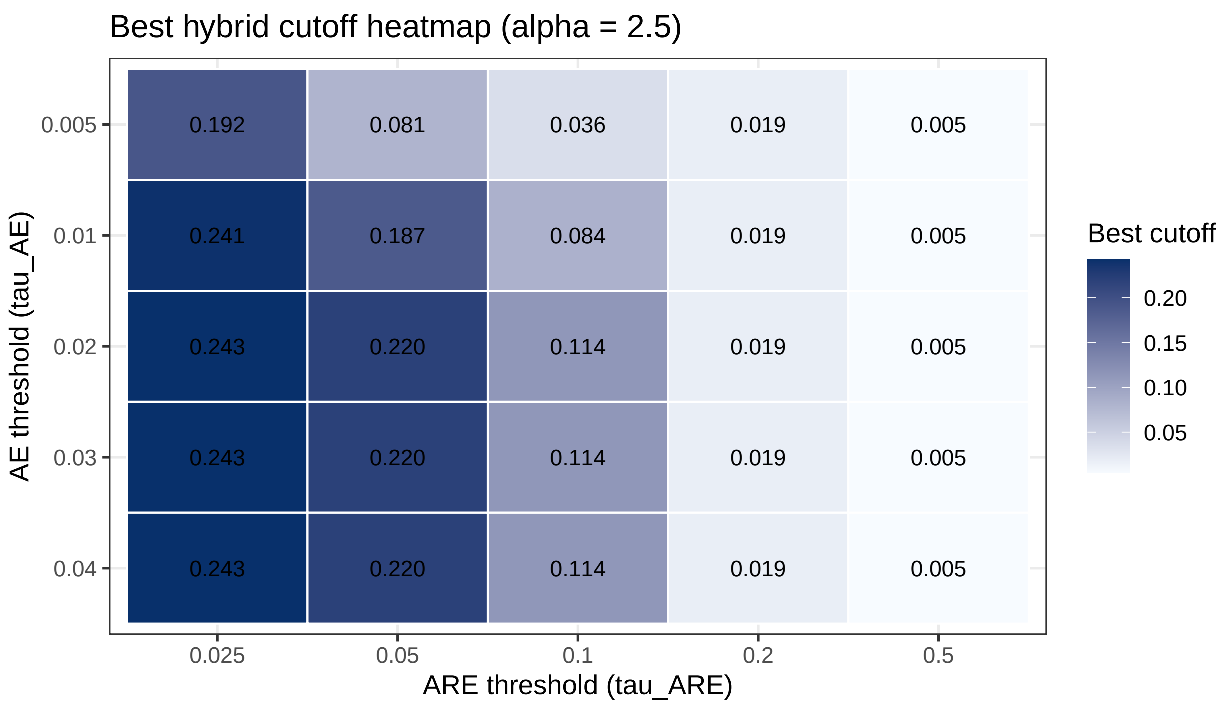

Figure 2: Best hybrid cutoff heatmap for alpha = 2.5.

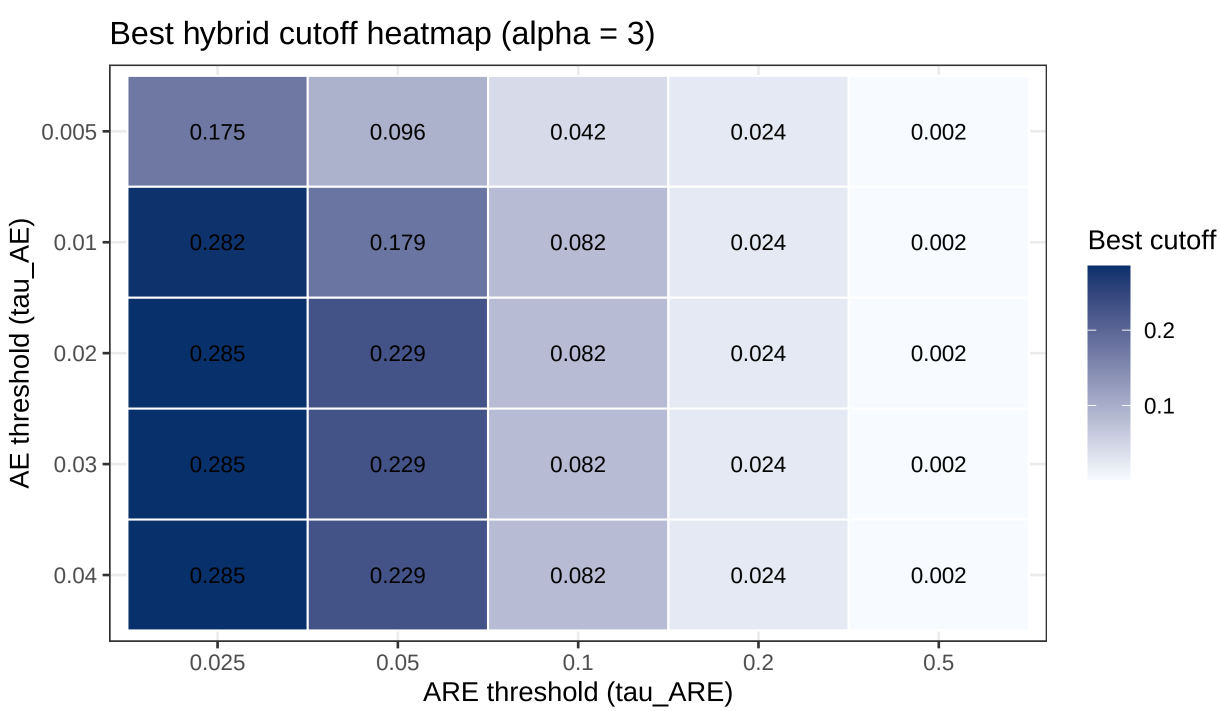

Figure 3: Best hybrid cutoff heatmap for alpha = 3.

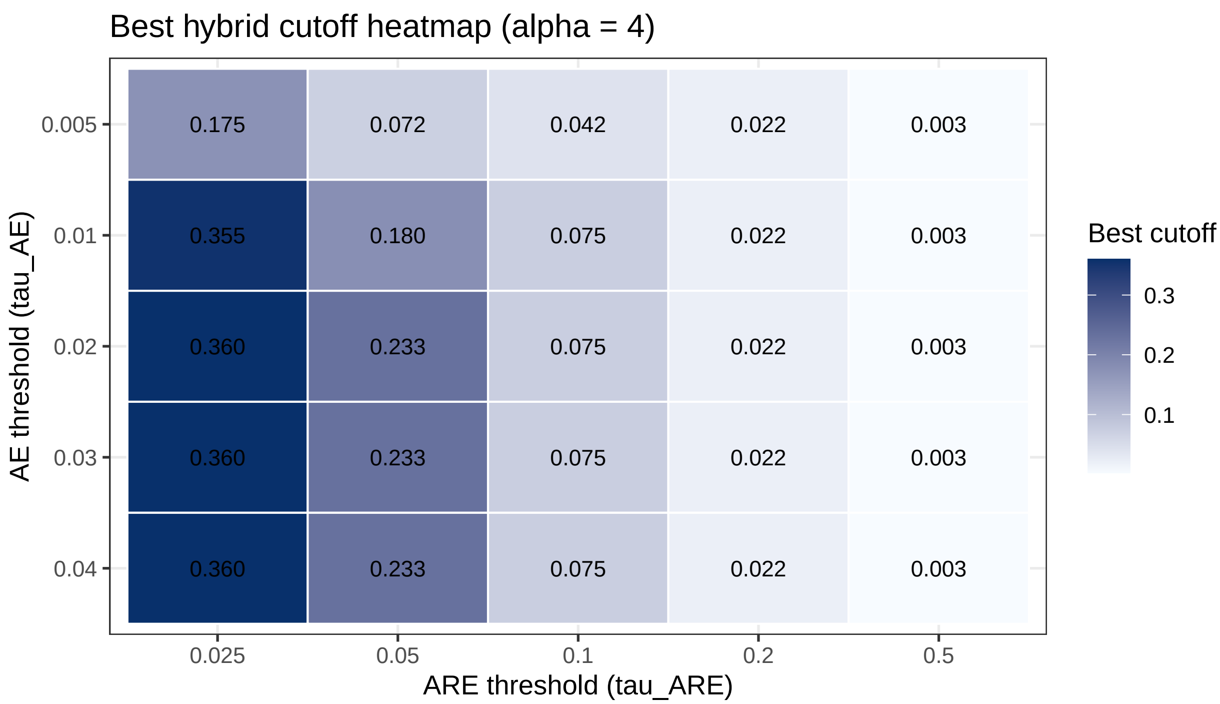

Figure 4: Best hybrid cutoff heatmap for alpha = 4.

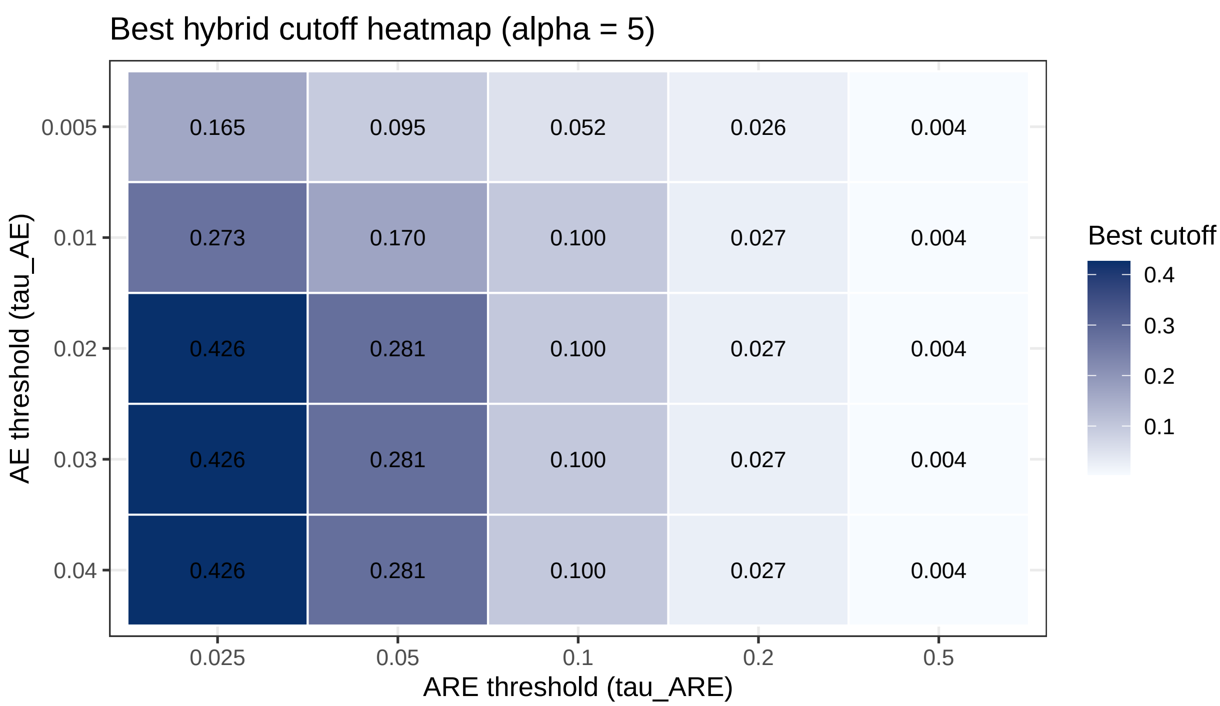

Figure 5: Best hybrid cutoff heatmap for alpha = 5.